

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality dropouts from canopy shade, darkness, or thermal crossover, frame-level detectors produce transient false positive spikes. In autonomous flight, each false alarm triggers wasteful loiter maneuvers, battery depletion, and spurious telemetry over bandwidth-limited Internet of Things (IoT) satellite or cellular links. This paper evaluates lightweight causal temporal decision methods for multimodal aerial SAR alert generation. Using a controlled 3×2 experimental design across three biomes (arid desert, temperate forest, and snow-covered alpine terrain) and two modalities (RGB and thermal), we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 GFLOPs at 640×640) trained in single-precision floating-point (FP32) arithmetic. Detection confidences are integrated via soft disjunctive late fusion and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark five temporal decision strategies under validation-optimized thresholds: baseline raw thresholding, moving average, median filtering, history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM). The methods are evaluated across positive event detection, false alert suppression, decision latency, and computational overhead.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) provide rapid wilderness search-and-rescue (SAR) aerial coverage [1], [2], but onboard vision systems operate under strict size, weight, power, and computing (SWaP-C) limits. Remote operations rely on low-bandwidth satellite or cellular Internet of Things (IoT) telemetry [3]. Spurious alerts trigger wasteful hover-and-verify loiters that drain battery reserves and dispatch rescue crews across hazardous terrain.

Wilderness optical sensing exhibits modality-specific failures. Visible red-green-blue (RGB) cameras degrade under dense canopy shadows and darkness, while thermal infrared (TIR) cameras detect human heat signatures regardless of illumination [4], [5]. However, thermal sensors suffer crossover

when terrain temperatures equilibrate with human skin radiation [6]. Heated rocks and clutter trigger false alarms, necessitating complementary dual-spectrum sensing. Snow-covered alpine terrain introduces a further failure mode: high-albedo snow cover can overexpose or wash out visible-spectrum imagery, while insulated cold-weather clothing can camouflage the thermal signature that would otherwise separate a person from a sub-zero background.

Edge aerial platforms rely on compact single-stage detectors such as You Only Look Once (YOLO) [7], [8]. We deploy Ultralytics YOLO11n [9] (2.6M parameters, 6.5 GFLOPs at 640×640) trained in single-precision floating-point (FP32) arithmetic. However, raw frame detections exhibit high volatility from platform vibration and clutter spikes, causing false alarms on empty scenes and transient dropouts on targets.

To isolate temporal dynamics, we structure a controlled 3×2 benchmark across three biomes (arid desert, temperate forest, snow-covered alpine terrain) and two modalities (RGB, thermal). We train two upstream YOLO11n detectors and fuse detections via soft disjunctive late fusion. Downstream, causal post-processing filters confidences across a five-frame sliding window (208 ms at 24 Hz). We benchmark five temporal filters, including a learned Mamba selective state-space model (SSSM) [10], under validation-calibrated thresholds.

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable alerts from multimodal detections for autonomous aerial search and rescue?*

This work delivers five main contributions: (1) **Corpus:** Dual-stream RGB/thermal benchmark across desert, forest, and snow-covered alpine biomes with positive targets and negative distractors. (2) **Baselines:** Standardized YOLO11n models (2.6M params, 6.5 GFLOPs at 640×640) trained in FP32. (3) **Decision Benchmark:** Empirical evaluation of five causal filters (raw thresholding, moving average, median, consensus, Mamba-SSSM) under validation-calibrated thresholds. (4) **Sensitivity:** Systematic evaluation across modalities and biomes. (5) **Resource Modeling:** Downstream flight endurance and IoT bandwidth impact analysis.

Section II reviews related literature; Section III details the MMSAR pipeline; Section IV outlines the experimental setup; Section V evaluates detection benchmarks, operational impacts, and limitations (Section V-D); and Section VI concludes

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

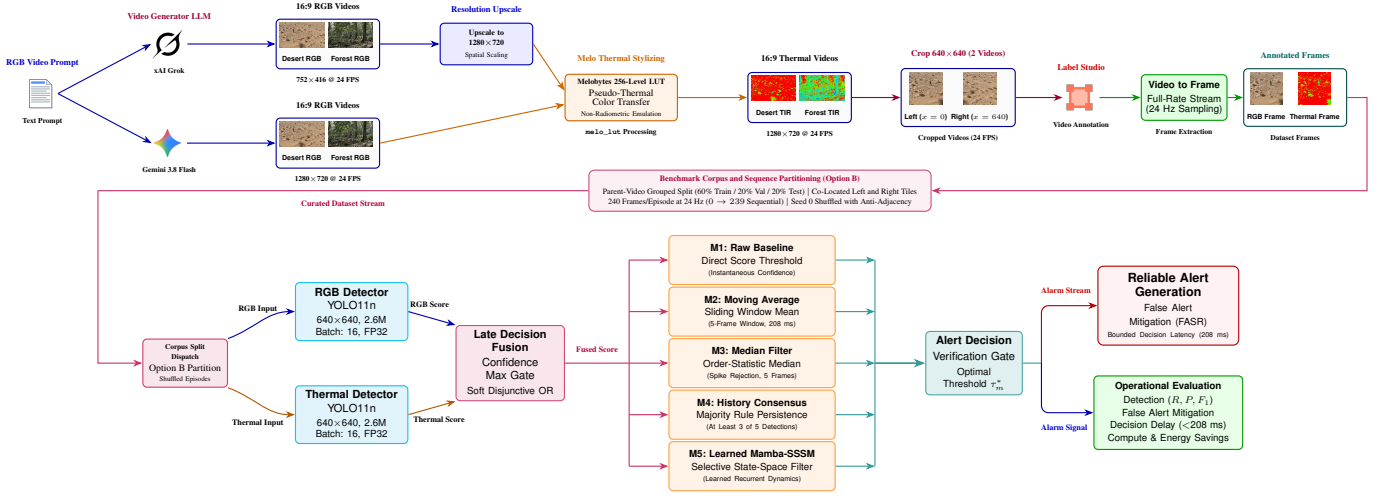


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB video streams are synthesized from text prompts using multimodal LLMs (Gemini 3.8 Flash and xAI Grok with resolution upscaling to 1280×720), followed by Melo thermal stylizing via a 256-level lookup table (`melo_lut`) to emulate pseudo-thermal infrared streams. The dual streams undergo dual-crop (640×640) spatial extraction, video-level annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under Option B (parent-video grouped split) with anti-adjacency snippet shuffling to eliminate cross-tile spatial leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), integrated via soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across five lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

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II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral sensing combines complementary spectral bands to overcome illumination extremes. Hwang et al. [4] introduced the KAIST benchmark, while Jia et al. [5] released LLVIP for visible-thermal surveillance. Liu et al. [11] showed that late fusion preserves modality features better than early concatenation. Li et al. [12] explored illumination weighting, and Zhang et al. [13] applied cross-modality attention. We adopt soft disjunctive late fusion via max-pooling, enabling decoupled detector execution without synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision models on UAVs requires balancing detection accuracy against SWaP-C constraints. Rudol and Doherty [1] demonstrated dual-band person geolocation for aerial search, while Silvagni et al. [2] showed that battery capacity bounds mission endurance. Lygouras et al. [14] showed that heavy ensembles exceed embedded envelopes, and Portmann et al. [6] identified thermal crossover as a major false alarm source. Kyrkou and Theocharides [3] addressed telemetry bottlenecks with EmergencyNet. We deploy Ultralytics YOLO11n [9] (2.6M parameters, 6.5 GFLOPs), building on single-stage detector evolution from YOLO [7], YOLOv7 [8], YOLOv9 [15], and YOLOv10 [16].

C. Temporal Post-Processing and Sequential Decision Making

Per-frame detectors applied to video produce high-frequency confidence chatter. While video tubelet methods such as Seq-NMS [17] and T-CNN [18] associate spatial

boxes across time, optical flow overhead is prohibitive on edge UAVs. Scalar temporal filtering avoids spatial tracking complexity. Classical sequential detection traces to Wald’s sequential probability ratio test [19], while radar signal processing employs sliding-window M -out-of- N binary integration [20]. Rabiner et al. [21] established the noise-rejection properties of running median and moving average filters. We adapt these causal principles to five-frame confidence streams under validation-calibrated thresholds.

D. State-Space Sequence Models

Recurrent networks suffer vanishing gradients, while Transformers scale quadratically. Structured state-space models (SSMs) provide linear sequence scaling with recurrent execution [22]. Gu and Dao [10] proposed Mamba, introducing data-dependent selection mechanisms with hardware-efficient parallel scans. Vision adaptations include Vision Mamba [23] and VMamba [24], with theoretical foundations formalized by Dao and Gu [25]. We adapt a pure-PyTorch selective SSM (Mamba-SSSM) as a causal temporal post-processor to filter environmental noise spikes.

III. METHODOLOGY

The MMSAR pipeline integrates dual-spectrum detection, soft disjunctive late fusion, and causal temporal filtering (Fig. 1). Synchronized visible RGB frames $I_{\text{RGB}}[n]$ and thermal frames $I_{\text{Thermal}}[n]$ produce per-frame confidences $c_{\text{RGB}}[n], c_{\text{Thermal}}[n] \in [0.0, 1.0]$.

A. Video Ingestion and Real-Time Budget

Autonomous UAV companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W TDP) require real-time processing matching sensor rates. Video streams are processed at

Algorithm 1 MMSAR Causal Alarm Decision (per frame n)

Require: RGB frame $I_{\text{RGB}}[n]$, thermal frame $I_{\text{Thermal}}[n]$
Ensure: Binary alarm $y[n] \in \{0, 1\}$

- 1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$
- 2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$
- 3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Late fusion gate}
- 4: $\mathcal{W}[n] \leftarrow \{s[n-4], \dots, s[n]\}$ {Causal window ($W = 5$)}
- 5: $y[n] \leftarrow \mathcal{M}_m(\mathcal{W}[n]; \tau_m^*)$ {Decision method m }
- 6: **if** $y[n] = 1$ **then**
- 7: Emit telemetry packet (S_{pkt}) and trigger UAV loiter maneuver
- 8: **end if**
- 9: **return** $y[n]$

$f_s = 24$ Hz ($T_s \approx 41.7$ ms). Frame-synchronous execution without packet drops requires satisfying:

$$\max(T_{\text{RGB}}, T_{\text{Thermal}}) + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (1)$$

where T_{RGB} and T_{Thermal} denote detector latencies, T_{fusion} is max-pooling latency ($O(1)$), and T_{postproc} is filter latency (sub-millisecond across all evaluated methods). Upstream detectors deploy two YOLO11n models [9] (2.6M parameters, 6.5 GFLOPs at 640×640): (1) **YOLO11n-RGB** and (2) **YOLO11n-Thermal**. Non-maximum suppression (NMS) failures yield confidence 0.0.

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality occlusions from causing dropouts, detector confidences are integrated via soft disjunctive late fusion:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (2)$$

This max gate ensures that confident detections in either spectrum propagate forward without cross-modality feature alignment. Algorithm 1 details the per-frame causal alarm pipeline.

C. Causal Temporal Post-Processing

Direct thresholding of raw confidence yields high false-alarm rates due to isolated spikes. Downstream processing filters the confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at $f_s = 24$ Hz): $\mathcal{W}[n] = \{s[n-4], \dots, s[n]\}$. Each method $m \in \{\text{M1}, \dots, \text{M5}\}$ is assigned an optimal decision threshold τ_m^* calibrated on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (3)$$

sweeping τ with step $\Delta\tau = 0.02$ across 46 candidate thresholds. Calibrated thresholds remain frozen for test evaluation.

We benchmark five causal post-processing methods: (1) **M1 (Baseline Raw)**: Instantaneous thresholding $y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*)$ ($O(1)$ time, 0 B state). (2) **M2 (Moving Average)**: Causal arithmetic mean $\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k]$, asserting $y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*)$ ($O(1)$ amortized, 20 B buffer). (3) **M3 (Median Filter)**: Order-statistic filter $\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n-4])$, asserting $y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*)$ ($O(W)$, 20 B buffer). (4) **M4 (History Consensus)**: Majority gate ($M = 3, W = 5$): $y_{\text{hist}}[n] =$

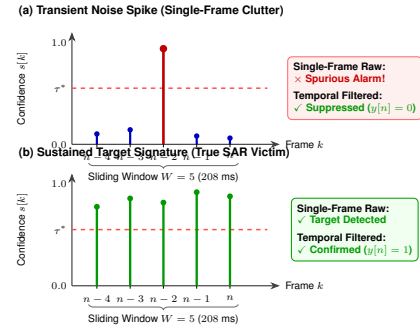


Fig. 2. Temporal signal behavior ($W = 5$ frames ≈ 208 ms at 24 Hz): (a) an isolated noise spike on a negative distractor scene triggers a spurious raw detection but is rejected by causal filtering ($y[n] = 0$), whereas (b) a sustained victim signature confirms target presence and asserts the alarm ($y[n] = 1$).

$\mathbb{I}(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq 3)$ ($O(W)$, 5 B buffer). (5) **M5 (Learned Mamba-SSM)**: Pure-PyTorch selective SSM ($d_{\text{model}} = 16, d_{\text{state}} = 8$): input projects via SiLU $x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}})$, yielding $\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$, $B_t = W_Bx_t$, $C_t = W_Cx_t$. Discretized dynamics $\bar{A}_t = \exp(-\Delta_t \exp(A_{\log}))$, $\bar{B}_t = (\Delta_t B_t) \odot x_t$ update hidden state $h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t$, asserting $y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(W_{\text{out}}u_t + b_{\text{out}}) \geq \tau_{\text{Mamba}}^*)$ (512 B state, 753 parameters).

Statistical confidence is established via sequence-level bootstrapping ($B = 1000$ iterations across 12 test clips), reporting 95% percentile confidence intervals (Table I). Fig. 2 illustrates transient noise rejection versus sustained confirmation.

D. Downstream Operational Metrics

Confirmed alarms ($y[n] = 1$) trigger telemetry packets ($S_{\text{pkt}} = 1.2$ kB standard, 45 kB thumbnail) over IoT links [3], requiring loiters ($T_{\text{loiter}} \in [15, 30]$ s) consuming $P_{\text{hover}} \approx 280$ W. Across K_{FP} false alarms, wasted bandwidth and energy are:

$$\Omega_{\text{wasted}} = K_{\text{FP}} \cdot S_{\text{pkt}}, \quad E_{\text{loiter}} = K_{\text{FP}} \cdot P_{\text{hover}} \cdot T_{\text{loiter}}. \quad (4)$$

False Alarm Suppression Ratio (FASR) evaluates relative reduction:

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (5)$$

IV. EXPERIMENTAL SETUP AND DATASET

Wilderness SAR dual-spectrum datasets remain scarce. As summarized in Table II, our controlled 3×2 design balances modality and environment. We evaluate across three wilderness biomes: (1) **Arid Desert**, characterized by rocky terrain, sparse vegetation, and midday thermal crossover where terrain temperatures equilibrate with human skin radiation; (2) **Temperate Forest**, characterized by dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors; and (3) **Snow-Covered Alpine**, characterized by high-albedo snow cover causing visible-spectrum overexposure and glare, sub-zero ambient backgrounds that elevate thermal contrast for exposed skin, and thermal camouflage from insulated cold-weather clothing that can suppress that same contrast. Each

TABLE I
COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER VALIDATION-OPTIMIZED THRESHOLDS ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Method	Time / Frame	State Mem	Val τ^*	Precision	Recall	F1-Score	F1 (95% CI)	Pos. Recall (%)	Negative FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	$O(1)$	0 B	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Five-Frame Moving Average	$O(1)$ amortized	20 B	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Five-Frame Median Filter	$O(W)$	20 B	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Five-Frame History Consensus	$O(W)$	5 B	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Learned Mamba-SSSM	$O(d_{\text{model}}d_{\text{state}})$	512 B	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

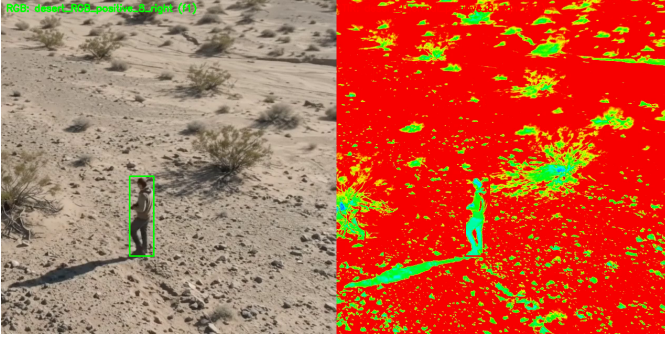


Fig. 3. Representative 2×1 co-registered detection viewport from the arid desert benchmark corpus (desert_RGB_positive_8_right, frame 0) fitting within a single column: visible RGB viewport (left) and synchronized Melo LUT pseudo-thermal infrared viewport (right) with ground-truth target bounding box annotation. [TODO: add a matching viewport figure for the snow/alpine biome once that corpus is finalized.]

environment comprises positive clips containing human targets and challenging negative distractor clips (sun-heated boulders, moving shadows, tree stumps, snow-covered rock outcrops, and unpopulated terrain).

Raw aerial video is recorded in 1280×720 format at 24 FPS. To preserve ground sampling distance (GSD) without downsampling small targets, a $2 \times$ dual-crop extraction harvests disjoint 640×640 snippets (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, $y = 40$). This produces 20 desert sequences (10 positive, 10 negative; 4,800 frames), 24 forest sequences (14 positive, 10 negative; 4,620 frames), and [PLACEHOLDER: N snow/alpine sequences (X positive, Y negative; Z frames) — pending completion of snow-biome capture/generation], totaling [PLACEHOLDER: total video streams and total frames at 24 Hz]. Video containers are remuxed with `-movflags +faststart` for streaming in Label Studio, where keyframe annotations with linear track interpolation yield ground-truth boxes. Negative frames are stored as empty 0-byte label files. Fig. 3 illustrates a representative co-registered 2×1 viewport pair.

A. Sequence Partitioning and Detector Training

Partitioning is conducted strictly at the sequence level under Option B (Parent-Video Grouped Episodic Partitioning) to prevent temporal data leakage. Derived sibling tiles from identical parent captures are co-located in the same partition. Sequences follow a 60/20/20 split: 60% training ([TBD] clips, [TBD] frames), 20% validation ([TBD] clips, [TBD] frames), and 20% testing ([TBD] clips, [TBD] frames). Clip frames

TABLE II
CORPUS PARTITIONING AND DETECTOR TRAINING SPECIFICATIONS

Parameter	Specification / Value
Environments	Arid Desert (20 clips), Temperate Forest (24 clips), Snow/Alpine ([TBD] clips)
Modalities	Synchronized visible RGB and thermal infrared (TIR)
Raw Flight Video	1280×720 at 24 FPS ($T_s \approx 41.7$ ms), 10.0 s nominal
Dual-Crop Snippets	$2 \times$ square tiles (640×640 , left/right, $y = 40$)
Corpus Frame Volume	[TBD] clips, [TBD] frames ([TBD] positive, [TBD] negative)
Sequence Split	Option B 60/20/20 ([TBD] train, [TBD] val, [TBD] test clips)
Upstream Detectors	Two YOLO11n (2.6M params, 6.5 GFLOPs at 640×640)
Training Setup	Batch 16, 100 epochs, FP32, SGD cosine ($\text{lr}_0 = 0.01$)
Decision Window	$W = 5$ frames (208 ms causal latency bound at 24 Hz)

TABLE III
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE (TEST SPLIT, $\tau = 0.50$)

Configuration / Condition	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-RGB (Temperate Forest)	[TBD]	[TBD]	[TBD]
YOLO11n-RGB (Snow/Alpine)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Temperate Forest)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Snow/Alpine)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Desert)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Forest)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Snow/Alpine)	[TBD]	[TBD]	[TBD]

TABLE IV
ENVIRONMENT-STRATIFIED FALSE-ALARM SUPPRESSION AND OPERATIONAL SAVINGS

Method	Biome	FASR	ΔFP	$\Delta \Omega$ (kB)	ΔE (kJ)	Δt_{fl} (min)
M2: Moving Avg.	Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Moving Avg.	Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Moving Avg.	Snow/Alpine	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median	Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median	Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median	Snow/Alpine	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Consensus	Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Consensus	Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Consensus	Snow/Alpine	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Snow/Alpine	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

remain in exact sequential order to support causal temporal modeling. Table II summarizes the training parameters.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Environments

We evaluate upstream frame-level detection of the two YOLO11n models on the test split. Table III summarizes precision, recall, and F1-score across modalities and environments under standard thresholding ($\tau = 0.50$).

Individual streams exhibit complementary trade-offs: visible RGB provides sharp spatial localization in illuminated desert and snow-exposed scenes but suffers dropouts under dense tree canopies and snow-glare overexposure; thermal infrared preserves target visibility through foliage occlusions but incurs false positive spikes on solar-heated terrain during midday desert passes and reduced separability against insulated cold-weather clothing in the alpine biome [PLACEHOLDER: *confirm empirically*]. Soft disjunctive fusion via max-pooling increases target recall by propagating detections from either spectrum forward. However, because max-pooling propagates the highest score, raw fusion also passes transient false alarm spikes from both sensors, necessitating causal temporal filtering.

B. Comparative Evaluation of Causal Post-Processing Methods

The five causal post-processing methods are evaluated on the fused confidence streams across the test split under optimal decision thresholds τ_m^* calibrated on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (3). Table I reports the comparative benchmark. The post-processing methods exhibit distinct filtering dynamics: (1) **M1 (Baseline Raw)**: Lacking temporal memory, memoryless thresholding allows isolated single-frame distractor spikes to trigger false alarms. (2) **M2 (Moving Average)**: Attenuates single-frame spikes by factor $1/W = 0.20$; calibrated threshold τ_{MA}^* adapts downward to compensate for smoothed target onset edges. (3) **M3 (Median Filter)**: Eliminates noise bursts under 3 frames while preserving sharp step edges as targets enter view. (4) **M4 (History Consensus)**: Requires at least 3 of 5 frames to exceed threshold τ_{hist}^* , enforcing discrete temporal consensus against transient clutter. (5) **M5 (Mamba-SSSM)**: Selectively dissipates isolated spikes through state decay while integrating persistent target motion into confirmed alerts with sub-millisecond per-frame latency.

C. Environmental Robustness and Downstream Operational Impact

Across biomes, distinct dynamics emerge: in the arid desert, sun-heated terrain induces thermal crossover spikes that temporal filtering attenuates; in the temperate forest, canopy occlusions cause intermittent RGB dropouts that late fusion and temporal state memory bridge; in the snow-covered alpine biome, high-albedo glare is expected to depress RGB confidence while thermal camouflage from insulated clothing may depress thermal confidence, a dual-modality stress case that late fusion alone cannot resolve and that should most clearly differentiate the five temporal decision methods [PLACEHOLDER: *confirm empirically*]. Suppressing spurious alarms prevents IoT telemetry queue congestion over low-bandwidth satellite links (2.4–10 kbps) [3]. Avoiding loiters ($T_{\text{loiter}} \in [15, 30]$ s at $P_{\text{hover}} \approx 280$ W) conserves onboard energy. Table IV models savings relative to M1: for each method m and biome e , false alarms avoided is $\Delta\text{FP}_m(e) = \text{FP}_{\text{baseline}}(e) - \text{FP}_m(e)$, saving bandwidth $\Delta\Omega_m(e) = \Delta\text{FP}_m(e) \cdot S_{\text{pkt}}$ ($S_{\text{pkt}} = 1.2$ kB standard, 45 kB

thumbnail), energy $\Delta E_m(e) = \Delta\text{FP}_m(e) \cdot P_{\text{hover}} \cdot T_{\text{loiter}}$, and adding flight endurance $\Delta t_{\text{flight}} = \Delta\text{FP}_m(e) \cdot T_{\text{loiter}}$.

D. Limitations

Several constraints bound our findings: (1) **Synthetic Corpus**: All video is generated by multimodal LLMs and stylized with a color lookup table to emulate thermal infrared; no real-world flown SAR footage or physical thermal-sensor imagery was used, and results should be read as characterizing temporal-filter behavior under controlled simulated conditions rather than validating a deployable detection system. (2) **Sample Size**: Evaluated on 12 held-out clips (2,520 frames at 24 Hz); larger flight campaigns are needed for wider generalization. (3) **Biome Diversity**: Constrained to desert, forest, and snow-covered alpine terrain; maritime and urban interfaces remain unmodeled. (4) **Flight Dynamics**: Sequences reflect nominal flight (30–50 m AGL at 8–15 m/s); turbulent sorties may alter jitter. (5) **Weather**: Clear conditions over static terrain, including static snow cover; dynamic weather events such as fog, rain, dust, and active snowfall/whiteout blizzard conditions require further study. (6) **Single Target**: Formulated for individual person search without multi-target disambiguation. (7) **Hardware Deployment**: Software-benchmarked; Jetson companion computer integration remains future work.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness SAR across desert, forest, and snow-covered alpine biomes using visible RGB and thermal infrared sensing. Dual YOLO11n baselines (2.6M parameters, 6.5 GFLOPs at 640×640 , FP32) were integrated via soft disjunctive late fusion. Five causal temporal post-processing methods across a five-frame window (208 ms at 24 Hz) were benchmarked under validation-calibrated thresholds. Causal temporal filtering effectively suppresses transient false alarms while maintaining target confirmation, conserving telemetry bandwidth and onboard battery reserves. Future work includes hardware-in-the-loop deployment on NVIDIA Jetson companion computers aboard physical UAVs and multi-target tracking.

DATA AND CODE AVAILABILITY

Simulation framework, snippet harvesting tools, and PyTorch temporal filters are archived at [PLACEHOLDER: code repository URL]. Video sequences and annotations are available at [PLACEHOLDER: dataset repository URL or DOI]. Pre-trained YOLO11n weights are deposited under [PLACEHOLDER: model weight repository URL].

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REFERENCES

- [1] P. Rudol and P. Doherty, "Human body detection and geolocation for UAV search and rescue operations using color and thermal imagery," in *Proceedings of the IEEE Aerospace Conference*, 2008, pp. 1–8.
- [2] R. Silvagni, A. Tonoli, E. Zenerino, and M. Chiaberge, "Multipurpose UAV for search and rescue operations in mountain avalanche events: Design considerations," *Geomatics, Natural Hazards and Risk*, vol. 8, no. 1, pp. 18–33, 2016.
- [3] C. Kyrkou and T. Theodoridis, "EmergencyNet: Efficient aerial image classification for drone-based emergency monitoring using atrous convolutional feature fusion," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 1687–1699, 2020.
- [4] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon, "Multispectral pedestrian detection: Benchmark dataset and baseline," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1037–1045.
- [5] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2021, pp. 3489–3497.
- [6] D. Portmann, S. Lynen, M. Chli, and R. Siegwart, "People detection and tracking from aerial thermal views," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 1794–1800.
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [8] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 7464–7475.
- [9] G. Jocher and J. Qiu, "Ultralytics YOLO11," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [10] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proceedings of the First Conference on Language Modeling (COLM)*, 2024, also arXiv:2312.00752.
- [11] J. Liu, S. Zhang, S. Wang, and D. N. Metaxas, "Multispectral deep neural networks for pedestrian detection," in *Proceedings of the British Machine Vision Conference (BMVC)*, 2016, pp. 73.1–73.13.
- [12] C. Li, D. Song, R. Tong, and M. Tang, "Illumination-aware faster R-CNN for robust multispectral pedestrian detection," *Pattern Recognition*, vol. 85, pp. 161–171, 2019.
- [13] L. Zhang, Z. Liu, S. Zhang, X. Yang, H. Qiao, K. Huang, and A. Hussain, "Cross-modality interactive attention network for multispectral pedestrian detection," *Information Fusion*, vol. 50, pp. 20–29, 2019.
- [14] V. Lygouras, N. Santavas, A. Taitzoglou, K. Tarchanidis, A. Mitropoulos, and A. Gasteratos, "Unsupervised human detection with an embedded vision system on a fully autonomous UAV for search and rescue operations," *Sensors*, vol. 19, no. 16, p. 3542, 2019.
- [15] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024, pp. 1–18, also arXiv:2402.13616.
- [16] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 107 982–108 006, also arXiv:2405.14458.
- [17] W. Han, P. Khorrami, T. L. Paine, P. Ramachandran, M. Babaeizadeh, H. Shi, J. Li, S. Yan, and T. S. Huang, "Seq-NMS for video object detection," *arXiv preprint arXiv:1602.08465*, 2016.
- [18] K. Kang, H. Li, T. Xiao, W. Ouyang, J. Yan, X. Liu, and X. Wang, "T-CNN: Tubelets with convolutional neural networks for object detection from videos," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 28, no. 10, pp. 2896–2907, 2018.
- [19] A. Wald, "Sequential tests of statistical hypotheses," *The Annals of Mathematical Statistics*, vol. 16, no. 2, pp. 117–186, 1945.
- [20] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.
- [21] L. R. Rabiner, M. R. Sambur, and C. E. Schmidt, "Applications of a nonlinear smoothing algorithm to speech processing," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 23, no. 6, pp. 552–557, 1975.
- [22] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022. [Online]. Available: <https://openreview.net/forum?id=uYLFoz1v1AC>
- [23] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual sequence modeling with bidirectional state spaces," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024, pp. 62 429–62 441, also arXiv:2401.09417.
- [24] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu, "VMamba: Visual state space model," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 103 031–103 063, also arXiv:2401.10166.
- [25] T. Dao and A. Gu, "Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024, pp. 10 041–10 071, also arXiv:2405.21060.